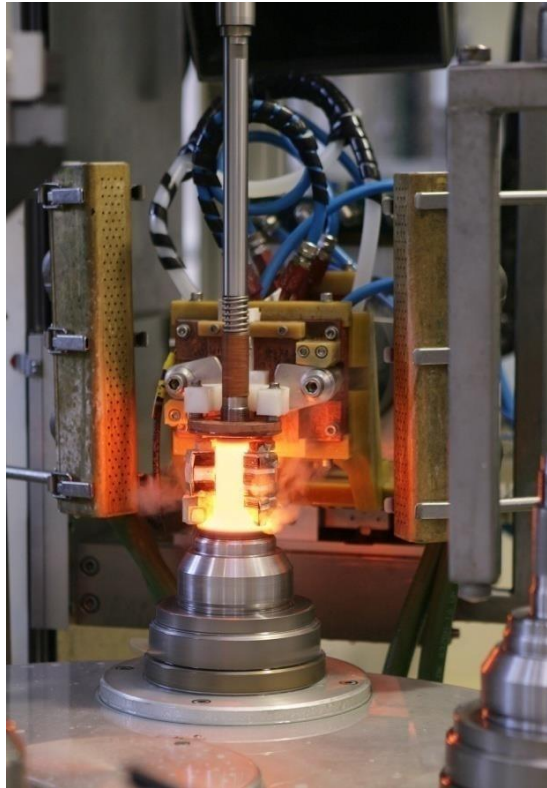


INDUCTION HARDENING MACHINE FORO/RACE AND T/HSG LONG STEM



Customer:

Contact:

Inquiry No:

Workpiece:

Production Plant:

HYUNDAI-WIA

Mr. Gyu-Tae Bae

WGT-M3-0021-003

Outer Races and Tulip housing Long stem

GH Induction India Pvt Ltd(India)

Your contact in Sales Dept.

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1. Records

1.1. Introduction

Following the request for quotation received from Hyundai-WIA for the supply of an induction hardening machine for outer races and tulip housing parts with final destination Chennai (India), and following the specifications.

This proposal is for Induction hardening of Outer race parts and of Tulip housing parts

The complete installation includes all the assemblies and devices required in the RFQ to follow completely its specifications for the treatment of O/Race and tulip housing parts for long stem.

GH Induction is proud to introduce HYUNDAI-WIA this proposal getting our best technical and service solution:

1. Huge experience on driveline heat treatment manufacturing with more than 300 references worldwide by GH group.
2. High-tech exclusive solutions for induction and specifically driveline applications.
3. Factory very near to Hyundai Wia India plant (about 30km). So, after sales support will be available on the same day.

For all this and many more reason we have worked to make sure that GH Group offers to HYUNDAI-WIA the best and safest technical solutions. We remain fully available to describe, in detail, our offer and answer to any query you may have.

1.2. Project requirements

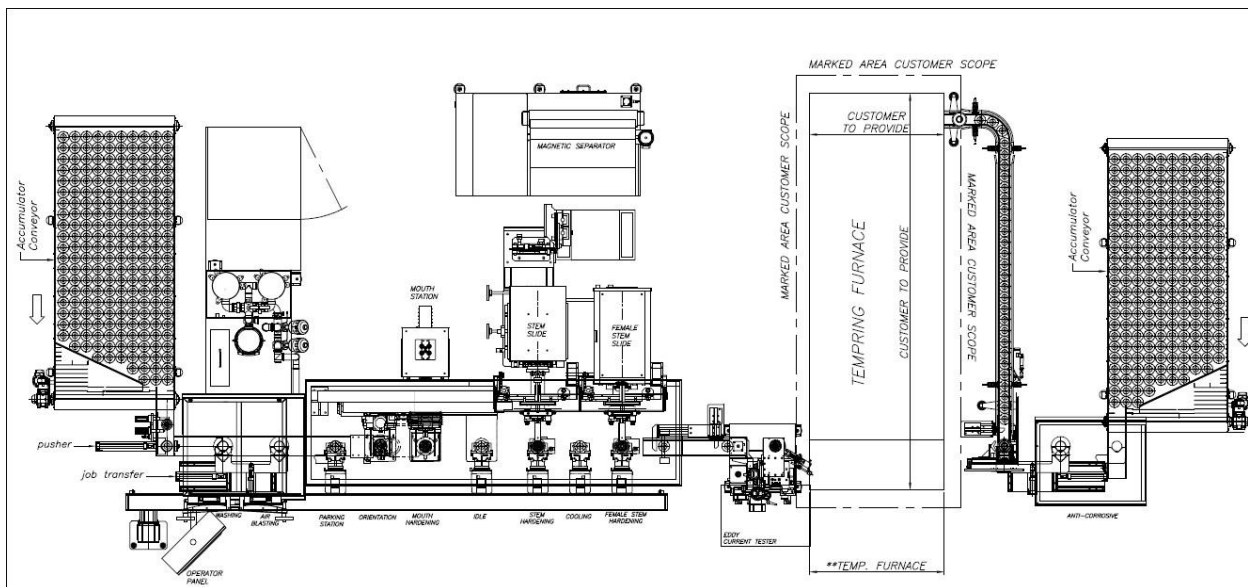
Part:			Outer Race and Tulip Housing for long stem		
List of parts:			As per the below chart		
NO.	SIZE	TYPE	Vehicle	Part Number	NO.
1	23	CWTJ	GPA	23023-GP901	1
2	23	CWTJ	GPA Long	23023-GP911	2
3	25	HDJ	GPA	21025-GP701	3
4	25	HDJ	GPA Long	21025-GP711	4
5	27	HDJ	GPA	21027-GP701	5
6	27	HDJ	GPA Long	21027-GP711	6
7	24	CWTJ	SION	21023-GP301	7
8	23	CTJ	BN7i	21025-GP901	8
9	23	CWTJ	BN7i	21027-GP901	9
10	24	CWTJ	KY	21023-H6001	10
Material:			XC55, XC48, S55CKN, Cf53		
Required cycle time:			29 sec		
Machine capability:			Max Ø 130 mm – Max Length 650 mm		
Surface hardness:			59~64 HRC after hardening		
Microstructure:			Bell: all Martensite / Stem: Perlite + Ferrite (Part of center Ø3)		
Deformations:			Serration O.P.D. max 0.02 mm		
Lay-out:			Within L8000mm x W6000mm x H 4500mm tentative		
Remarks:			The validity of the present proposal is not extensive to other parts or applications not specified here.		

121 1. Proposal

The whole installation includes the following circuit:

1. Input accumulator conveyor with buffer for up to 200 parts.
2. Washing and drying station.
3. Linear transfer is considered for Hardening machine in the following scheme:
 - 1) Positioning (for outer race it will be idle)
 - 2) Bell hardening.
 - 3) Idle
 - 4) Stem hardening.
 - 5) Auxiliary quench.
4. Tempering (to be supplied by Hyundai WIA)
5. Eddy current testing.
6. Laser marking station
7. Anti-corrosive station.
8. Output Accumulator conveyor with buffer for up to 200 parts.

For the hardening machine a common 500kW generator has been considered for bells and tracks (switched stations)



Tentative layout female station is eliminated and laser marking machine and station is included.

1.3. Production conditions

Availability

Cycle time 29"
 Production rate 95%

Installation site

Place, address Chennai(India)
 Temperature, min. / max, 0-45°C
 Relative humidity, max, ≤85%

Required supplies

Electricity 3 x 415 V -50 Hz
 Industrial water 20 ÷ 28°C; 3 ÷ 7 bar
 Compressed air 4 ÷ 6 bar

REMARKS:

- If the customer has no proper water utility circuit within his plant to fulfil required features, GH could provide consulting services, on the basis of your instructions, and propose alternative solutions.
- If the max inlet temperature is above 28°C, then a chiller unit with separate circuit for the converter is needed.

2. Scope of supply

The supplies related to the present proposal are:

2.1. Engineering		
Induction application parameters study and test	YES	
Equipment and project engineering	YES	
2.2. Manufacturing		
Components purchasing and equipment assembly	YES	
Assembly and put in operation	YES	
Technical documents	YES	
Pre-acceptance tests after manufacturing	YES	
Process statistic analysis &CPk ¹ verification before delivery		NO
2.3. Delivery		
Dismantling for transport and packaging	YES	
Transport and its assurances to final destination	YES	
Handling in destination and placing in customer's factory		NO
Link to the customer's mains (AC supply, water, pneumatics...)		NO
Reassembly and put into production conditions on customer's plant	YES	
Training of customer's personnel	YES	
Final acceptance tests on customer's site	YES	
Technical support to the starting of production (one week after SOP(one mechanical, one electrical)).	YES	
2.4. Standards		
Standard GH: Installations Manufacturing		
2.5. Technical documentation		
As itemized in Standard GH: Installations Manufacturing		
Documents	Five Hard copies+ soft copy by usb	

1. CPk: capability index intended to study the process repeatability and stability under normal work conditions

3. Equipment description

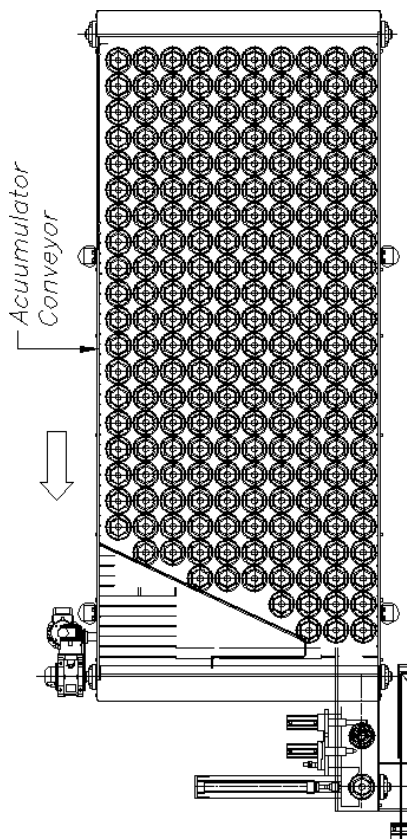
3.1. Input buffer and washing station

3.1.1. 1. Input buffer.

The conveyors are made by aluminium profiles / steel channels and stainless steel hinges. It has legs in aluminium profiles / steel channels to support on the floor. The conveyor is driven by a geared motor.

Input conveyor has been enlarged forming multiple rows in order to dispose a buffer capacity upto 200 parts prior to the washing station.

Fig.1 Diagram from input conveyor to washing station.

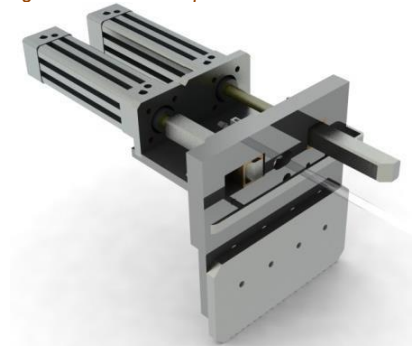


3.1.2. Pneumatic Dispenser

A singling unit is mounted before the washing station (at the end of the conveyor) to deliver one component at a time.

A manipulator with pneumatic clamps loads the parts from conveyor to the washing station in a horizontal movement.

Fig.2 Pneumatic dispenser.

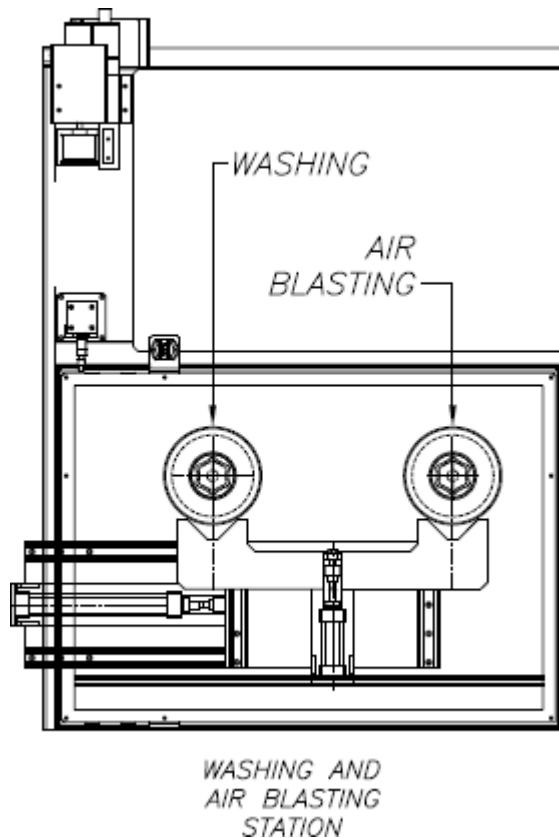


3.1.3. Washing Station.

Once the part is placed over the washing platform, the shower will come down and wash directly inside the bowl and outside, wash the part with hot water (30–60°C) under pressure.

After the washing, in next location an air blowing ring dries the part from top to bottom before transfer to hardening station.

Fig.3 Washing and drying station



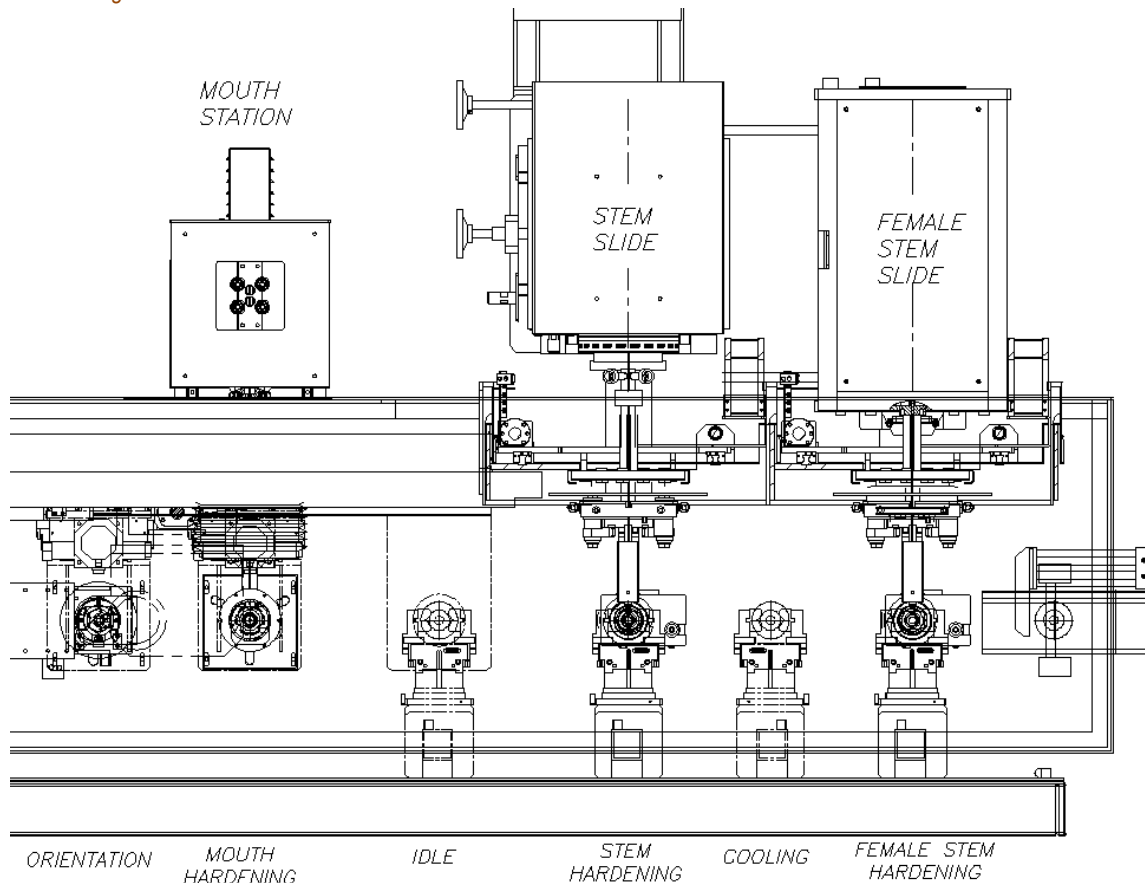
The washing station includes a stainless-steel tank of 500L with heat resistances, double bag filter and oil skimmer. Once the cycle is done the manipulator takes the part to the positioning station.

3.2. Hardening Machine

Linear transfer is considered for Hardening machine in the following scheme:

- 1) Positioning (for outer race it will be Idle location)
- 2) Bell hardening.
- 3) Idle
- 4) Stem hardening.
- 5) Auxiliary quench.

Fig.4 Hardening machine sketch



Tentative layout female OR not included and laser marking machine included.

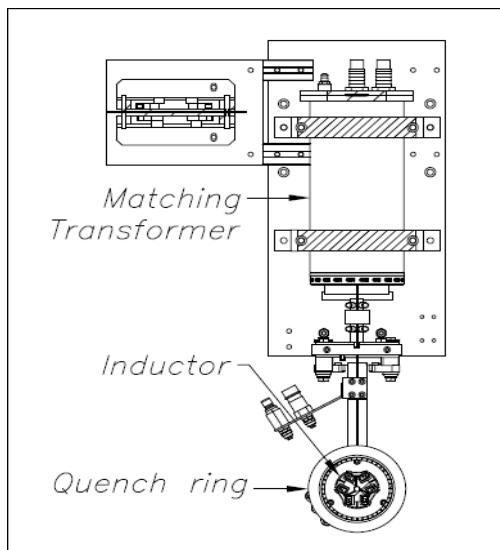
321 1. Positioning

The part will transfer from the washing machine by the pneumatic operated manipulator transfer with grippers. Here the part positioning will be done for the bell orientation. For outer race, it will be idle location.

322 2. Bell hardening

This station is having rotating chuck for hardening and transferring the part from orientation to bell hardening and then to idle location. This total assembly will move in X and Y direction by servo controlled motor. Chuck will pick the part from orientation location, harden the part in Hardening location and place it in the idle location. Again, goes to the orientation location to pick the next part. The chuck will have the LVDT to check the part length throughout the operation. This is to check the part slip during the operation. The bell hardening will be done in single shot method with rotation for outer race parts and scan hardening will be done for tulip housing without rotation.

Fig.5 Bell hardening station



323 3. Stem hardening

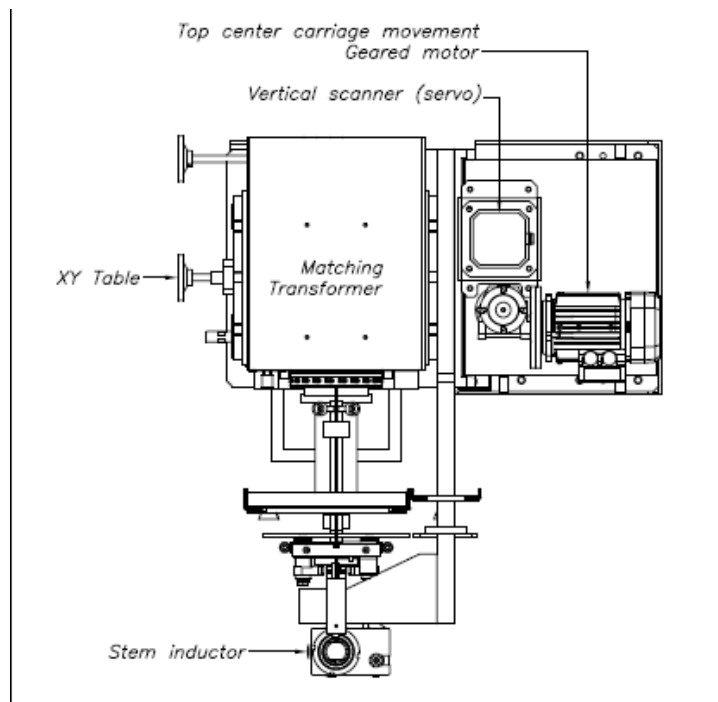
The manipulator will take the part from the idle location to stem hardening location. Here the stem single shot hardening will be done. The stem coil vertical movement will be by servo motor.

After stem hardening the part will be transferred to auxiliary quenching location. Here the part will get cooled. After completion of auxiliary quenching, All the XY manual table will have LVDT to monitor the XY position.

To avoid corrosion and degradation on elements in contact with the quenching water, the machine has been designed with an isolating wall which separates “Dry” and “Wet” areas.

So, splashes and vapours contact is minimized with critical components such as transformers, XY tables or ball screws placed in the dry zone which clearly enlarge their life.

Fig.6 Stem hardening station



3.3. Eddy Current testing

The part will be transferred to the eddy current station by the manipulator and eddy current testing will be done.

Eddy Current brand IBG model Eddy-Visor DS 2 channel checks bells and stem hardening and material. Once checked, the good parts are transferred to the dot matrix station by the manipulator.

3.4. Dot marking station

Parts are taken from the eddy current station by a pneumatic manipulator and part will be locked for marking by pneumatic clamp.

Dot marker JEIL MTECH Marksman MCU-100N Model marks the part at the described position on the bell.

Marked parts are transferred to the anti-corrosive station by a pneumatic gripper.

LASER MARKING MACHINE AND STATION INCLUDED

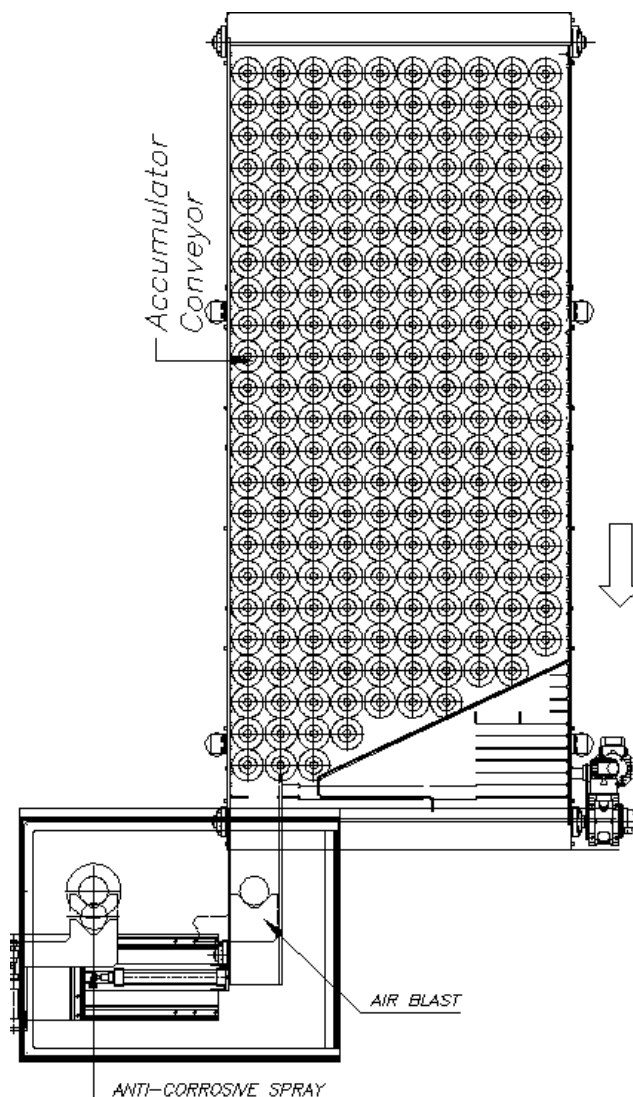
1 **Note: Noise level close to main pumps and dot marking station won't accomplish <75dB noise level requirement without an acoustic capsule isolation.*

3.5. Anti-corrosive station and output buffer

Once the part reached at this location, the anti-corrosive spray shower will come down and spray the oil over and inside the part.

In next station, the oil will be dried by air blow and the parts are left over the output conveyor which has been enlarged forming multiple rows in order to dispose a buffer capacity upto 200 parts.

Fig.11 Output buffer diagram.



3.6. Electric system, control and command devices

1. Control system

The main function of the control & command system is the machine running and managing always subordinated to the safety of personnel and environment.

The hardening machines have independent control systems.

Hardening machine is composed of:

Computer Numeric Control (CNC) Mitsubishi, with 5 axes servo controlled

Man - Machine Interface (MMI) Mitsubishi

Hand held - Machine Interface (HMI) Mitsubishi

Quality process supervision system

The operation concerned by this proposal is controlled by supervision of the following parameters:

MF/HF output frequency.....	YES	
MF/HF output power	YES	
Energy delivered during the heating.....	YES	
Heating time	YES	
Product cooling water flow.....	YES	
Product cooling water temperature	YES	
Quenching time.....	YES	
Product rotation.....	YES	
Coil position and/or speed	YES	
PMS - GH Process Graphic Control.....	YES	
Datalogger.....	YES	

Parameters supervision:

- Integrated in control & command system (CNC / PLC)
- Performed by independent system

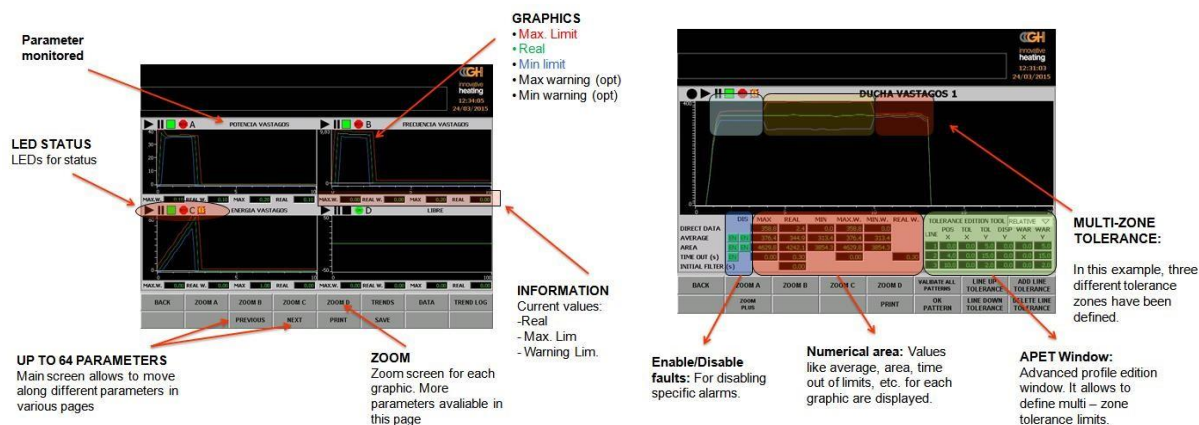
In case of non-conformity with the reference values of one or several parameters, the quality following system

Alerts the operator with an alarm signal	YES	
Starts the product automatic separation	YES	
Stops the machine	(User setting)	

Some screen examples of PMS - GH Process Graphic Control

GHPGC is the process monitoring system (PMS) developed by GH that allows to monitor and to control main heating process parameters in the safest and most functional way.

Fig.12 PMS screen examples.



It is carried out the graphic control of up to 99 parameters such as (*depending on the sensors and devices installed in machine):

Heating process	Cooling process	Hardening depth
Heating time	Quenching time	Frequency
Part energy	Quenching flow	Heating time
Power	Quenching temperature	
Voltage		
Current		
Frequency		

In order to evaluate if a process is valid, main parameters are recorded and plotted in real time according to sampling time during whole cycle. When a good part is validated, MAX and MIN limits are calculated and plotted for each parameter in accordance with the tolerance values defined by the customer. After this moment, parts produced are monitored and compared with the defined limits. If any parameter detected is out of the limits, an alarm is triggered, and part is considered as a NOK part.

2. Generalities

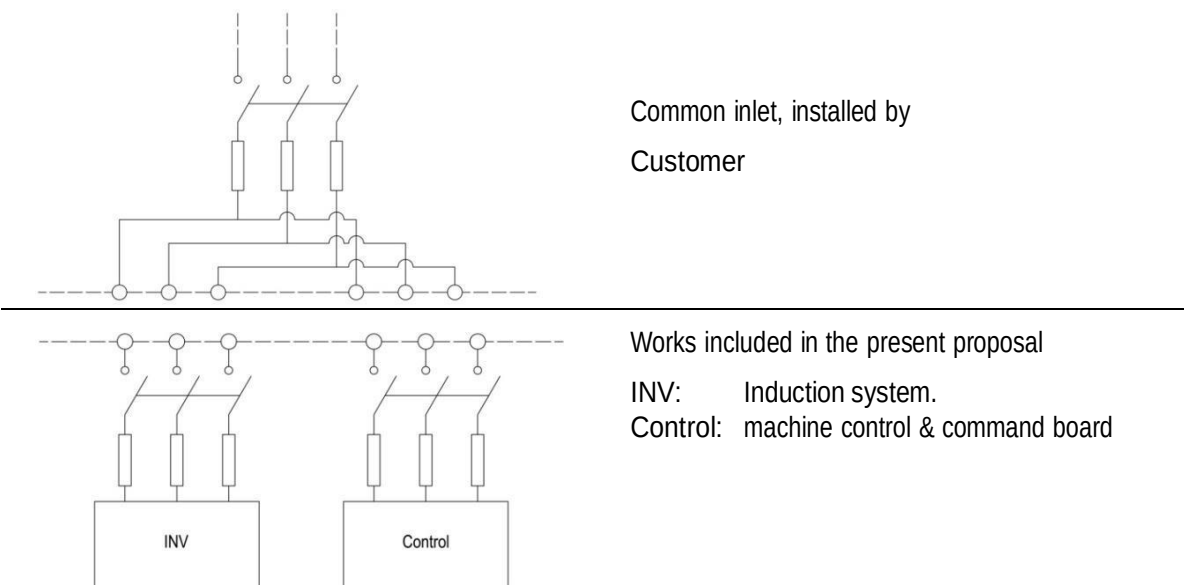
The electric system is made in accordance with standard GH NG 1.7

The execution of electric inlet includes two independent lines:

- One, for the induction equipment (generator and peripherals) and
- Other, for the general control system, mechanisms and electric devices in general.

This solution allow to keep a part working while the other is disconnected, functioning mode necessary for induction generator adjusting, maintenance and repairing operations.

This double inlet is compatible with an eventual common line, usually installed by the customer, with the usual protection, measure and/or breaking elements.



The protection grade of boards is:

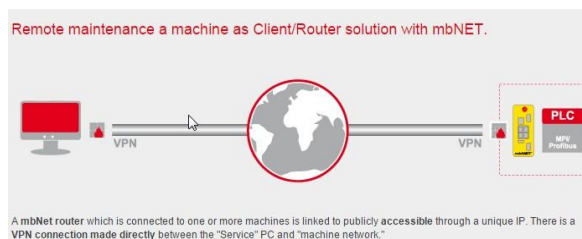
IP Material

54 Electric boards in general: generator, oscillating circuit, command panels

3.7. GH Remote assistance service

With the aim to assure a better service meanwhile a remote assistance service, GH installs a wireless router in the machine which allows a remote access to our technical service staff to solve possible incidences directly in the PLC from our own facilities in GH Induction India Pvt Ltd (India). This new tool would help us in a quicker answer or in case of any technical assistance.

The access to the machine is done through a secure VPN connection by a dedicated hardware (WAN or 3G connection). Internet SIM card is not included on supply.



This service will be given free of charge during all the warranty period. After this time, the customer may return the hardware to GH or buy it for INR 200,000.00 . The service fee for a 24h assistance during a year (after warranty period) will be of INR 250,000.00 and in case of technical assistance requirement the charge will be measured per hours (INR 2000.00/hour).

3.8. Transistors induction generator (Series)

381 1. Functioning

The induction generator is the system destined to convert the energy coming from the industrial AC mains in a medium or high frequency signal adapted to the feeding of the induction heating station. This signal is a square wave that goes through an oscillating circuit (capacitors bank and an inductor as a minimum) and generates a magnetic field. Once that a part is under this magnetic field, electrical currents are made upon it. These currents increase the superficial temperature of the part by means of the Joule effect.

To obtain the square signal, the AC industrial current is rectified to a DC current. After this, a transistors inverter bridge obtains the desired AC wave at the adequate frequency for the application.

382 2. Security

The ensemble formed by generator and oscillating circuit is water refrigerated. The equipment protection is granted by the Reading of the integrated flow-meters. The control electronics keeps always protected the semiconductors against internal and external failures (inductor breakage, short-circuits, overloads...).

GH generators accomplish with Machine Security Standard ISO 13849-1, category PL d

383 3. Technical data with tentative dimensions

The selected generators are 250SM20 for the hardening. The following chart shows the main features of SM generator series.

Type		12SM	25SM	50SM	75SM	100SM	150SM	200SM	250SM	300SM	400SM	500SM	600SM	800SM		
Output continuous power	kW	12	25	50	75	100	150	200	250	300	400	500	600	800		
Frequency	kHz	[0.5, 20] [20, 150]						[0.5, 20]								
Power supply	kVA	15	31,25	62,5	93,75	125	187,5	250	312,5	375	500	625	750	1000		
Voltage supply		380–480 VAC 50/60 Hz			380-480 VAC 50/60 Hz ⁽¹⁾											
Generator width	in/mm	23,6/600			31,5/800		47,2/1200		55,1/1400		55,1/1400		102,3/2600		118,1/3000	
Generator depth	in/mm	23,6-31,5/600-800			31,5/800		31,5/800		31,5/800		31,5/800		35,4/900		35,4/900	
Generator height	in/mm	43,3/1100			78,7/2000		78,7/2000		78,7/2000		78,7/2000		78,7/2000		78,7/2000	
Width with cooling system	in/mm	23,6/600			78,7/2000		78,7/2000		86,6/2200		90,5/2300		137,8/3500		157,5/4000	
Depth with cooling system	in/mm	23,6/600		31,5/800		31,5/800		31,5/800		31,5/800		35,4/900		35,4/900		
Height with cooling system	in/mm	74,5/1900			78,7/2000		78,7/2000		78,7/2000		78,7/2000		78,7/2000		78,7/2000	
Water temperature min/max	°F/°C	[68, 98,6] / [20, 37]														
Water supply		1"								1" ¼"			1" ½"			
Waterflow	gpm/lpm	2,6/10			3,9/15		7,9/30		8,8/40		12,7/48		18,5/70		30,38/115	36,7/140

384 4. Generator interface

The induction generator allows its control and command interface through Mitsubishi HMI. Its user-friendly interface makes able to supervise the equipment general functioning and to control the output signal.



Fig.13 Power supply screen examples:

A: Main screen

B: Flow-meters visualization

C: Process parameters configuration

D: Configuration screen

The interface allows the user to visualize the state of the equipment (prepared, heating, fail); visualize alarms (25 last events); configure the functioning parameters of the equipment. All this is done on an easy and intuitive way, "surfing" through different screens:

- Visualization of generator's analogical parameters: power, current, voltage, frequency and energy. Power regulation set.
- Screens related to the watch of flow-meters.
- Screens for parameters configuration and functioning mode.

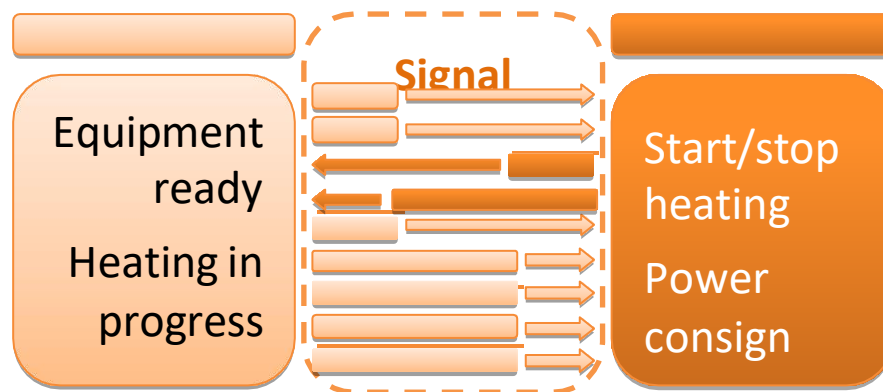
Optionally, not included on the base supply, the generator can store programs to implement temperature controlled automatic heating cycles.

Also optionally, not included on the base supply, the generator can work as the main control for a simple machine by means of an internal program system (IPS) developed by GH. This program must be done by GH personnel.

385 5. Connection to an external control system

The generator is also ready to be controlled from an external system, so that the induction heating equipment is integrated into a system controlled by a general device. In order to do this, it could be linked through bus connections for its external control by means of: Profibus, Profinet, Device-net, Interbus.

Optionally, the generator can be equipped with different analogical inputs / outputs destined to exchange the following signals:



3.9. Oscillating circuit

The capacitors bank is the element providing the necessary capacity to establish, with the self-inductance of the inductor eventually augmented by the impedance adaptation transformer when applied, the resonant output circuit which will define the frequency of the induction signal.

Essentially, the oscillating circuit consists of one or several capacitors chosen depending on the oscillating conditions to state.

The capacitors are built in mixed dielectric, biodegradable, free of any pollution danger. The permissible overload is 1,35 the nominal current, including harmonics.

The oscillating circuit includes a **coil interface transformer**. This is a device intended to increase the coil self-inductance, when this value becomes too low to compose an oscillating circuit with an appropriated level of power allowing the induction heating.

The working principle of this transformer is based on the multiplication of the coil self-inductance, L , by a value, n , given by the transformer ratio.

Depending on the required ratio, the working voltage, power and frequency to apply, the transformer may be manufactured under different versions but it is always made under an isolating and protection cover.

Constructive solution details

Capacitors bank with adjustable capacity	YES
Oscillating circuit under IP54 cover.....	YES
Oscillating circuit equipped with coil interface transformer	YES
Hardening quench circuit equipped with quick connection plugs	YES
Coil interface transformer equipped with adjusting taps	YES

The safety against electric risks for access to zones under voltage is guaranteed by:

- Occasional access zones: panels screwed or fixed by other elements needing tools for dismantling.
- Doors: lock with key plus opening detection with automatic disconnection of the power

3.10. Coil or inductor

The coil or inductor is the element in the induction heating system performing the direct action on the product to process, it means: it is the component destined to transfer the energy to the material to treat. With this aim, the inductor is built with the form and size appropriated to the product.

The coil is manufactured as an assembly rigid and stable against the mechanical or thermal efforts, allowing a repetitive and controlled operation and a free of errors placing and positioning.

The main parts are:

Coil: with a turns number and shape specially chosen to obtain maximum efficiency. The coil is built with copper pipe intended to operate as internal cooling circuit destined to evacuate the residual heat due to the Joule effect losses.

Fastening system: it works as electric connectors and as mechanical support and positioning.

Cooling circuit fitting: it is the water input / output elements connecting the coil to the coolant circuit.

The coil is equipped with a cooling quench destined to refrigerate the product after the action of the heating coil. This quench is built with magnetic inert materials in order to avoid any effect on inductor field.

Constructive solution details

Coil equipped with fast connecting system	YES
Coolant circuit equipped with fast connecting system	YES
Coil equipped with cooling quench.....	YES
Cooling quench equipped with fast connecting system	YES
Quench included in the machine.....	YES

Non-contractual examples of different coils to be applied on this machine:

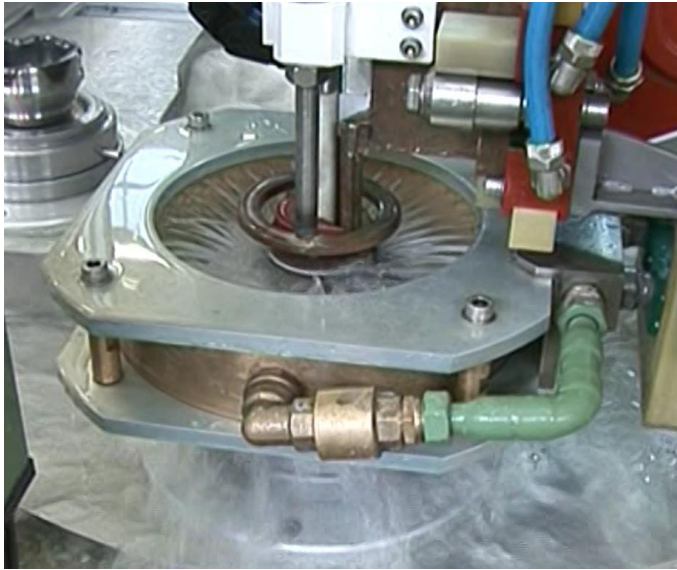


Fig. 14: Bells single-shot coil and quench set

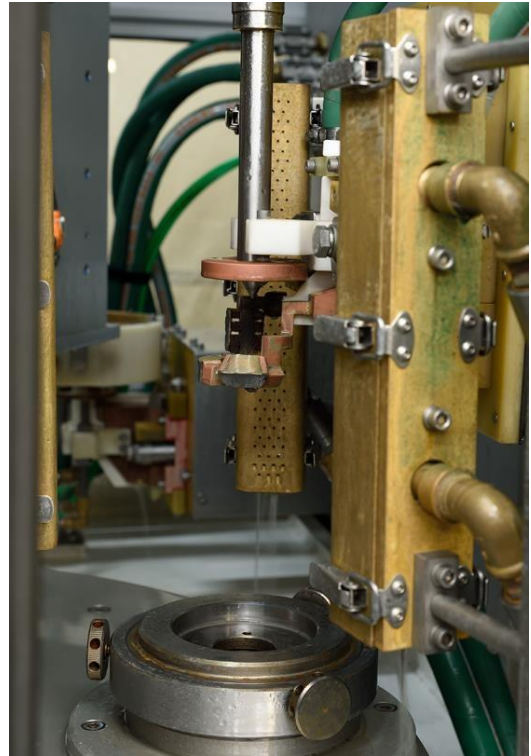


Fig. 15: Stem single-shot coil and quench set

301 1. 3DPCoil(optional not included in this offer)

GH Group has patented a new system for inductors design and manufacturing using exclusive 3D printing technology. Results in comparison with the traditional manufacturing process are:

- **Reduction of costs per part:** 3DPCoils have a longer working life thanks to its solid design in one single part free of welding's, which revert to a highly increased productivity. Additionally, to this, as their geometrical receptivity is guaranteed, its maintenance and recalibration time are minimized.
- **Reduction of stocks:** The increase in cycles of the inductors combined with the reduction in delivery time result in a reduction of spare stock.
- **Savings in maintenance:** The increase in cycles reduces the consumption of inductor spares. Furthermore, the shorter reference changes minimize the labour time required for such changes.
- **Quality of the parts:** The ability to manufacture and supply identical coils ensures that heating profiles are maintained when fixtures are changed. This technology allows also complex coil's designs impossible to conceive by handcraft manufacturing.



GH 3DPCoil (patent) printed in one unique part.



GH Microfusion (patent) inductor with embedded quench.

GHbased on the application will decidewhat inductor's technology may be used.

3.11. Induction heating cooling system

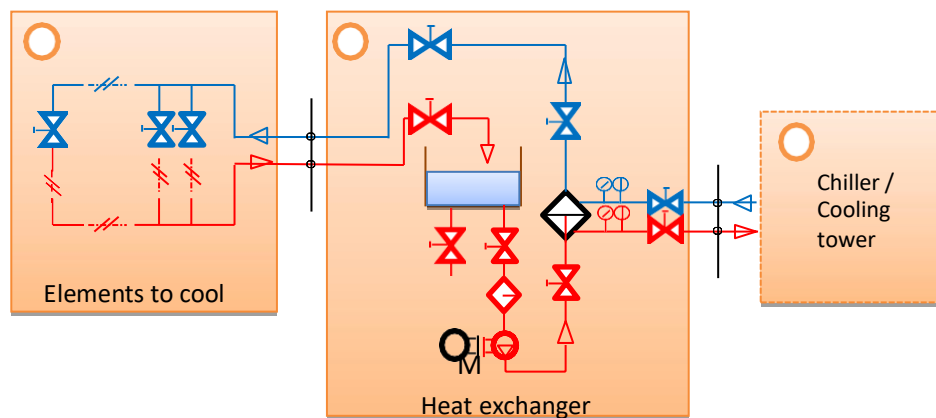
Some elements working in the induction heating system (e. g., transistors, capacitors, inductors) under the action of high currents require cooling to evacuate the heat by Joule effect.

This cooling is based on a closed loop of liquid coolant through these elements. The system is named by GH as R3.

The liquid coolant applied in this circuit is demineralised² water. The low content of salts reduces the risk of obstruction by solids.

The coolant itself is cooled in a heat exchanger, where it transfers the heat to a second coolant, usually industrial water (water without any special chemical requirements).

Finally, the industrial water is cooled in a cooling tower or in a frigorific machine or chiller.



These three sections are independent systems:

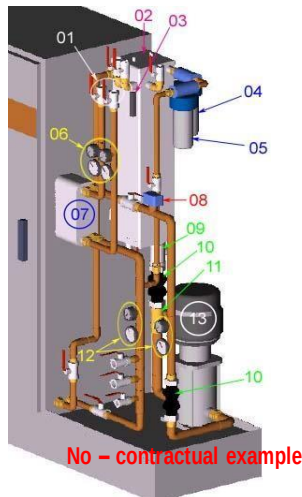
1. The coolant circuit is integrated in the induction system and it is composed of a circuit with its appropriated control and measurement devices (valves, flow meters, thermostats...). Its function is to conduct the coolant through the components to cool in order to guarantee the induction system working temperature. In case of lack of capacity (due to coolant failure or excessive temperature), the integrated protections stop the equipment under failure condition.
2. The R3 unit is a system intended to maintain the coolant temperature as a thermal interface between the coolant circuit and the industrial water network
3. The external cooling equipment (usually a cooling tower or a chiller) is a part of the industrial water network not included in the induction heating equipment.

See in the chapter concerning the scope of supply the parts included in the present proposal.

2. Avoid all confusion with distilled water which is not appropriated for this purpose because it acts as an acid due to its tendency to ionization under the temperature and voltage conditions present in the cooling circuits and in contact with the metallic parts of such circuits. The demineralized water is practically inert, dielectric and non-polluting (no risks are associated to the lose)

3.11.1. R3 unit – demineralised water cooling system

The cooling R3 system is a well proven standard design of GH, fitted into most GH machines and generators. Treated water is used as coolant for the equipment in a closed circuit and is cooled at its turn by industrial water as secondary coolant in open circuit. The heat exchanger is of plate type.



- 01 : Industrial water input/output
- 02 : Upper access to the cooling water tank
- 03 : Level indicator
- 04 : Polypropylene filter
- 05 : Deionising resin vase
- 06 : Thermometer & manometer
- 07 : Heat exchanger
- 09 : Output valve
- 10 : Attenuating connectors
- 12 : Thermometer & manometer
- 13 : Pump

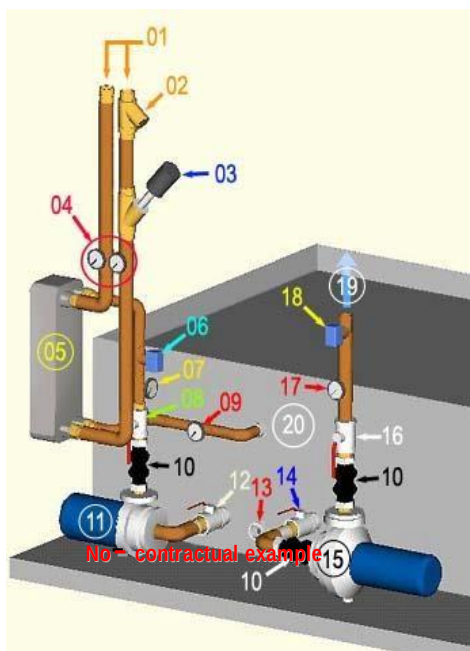
3.12. Hardening liquid cooling circuit

The product cooling water is applied on the product by quenching through a specific quench. The flow and temperature are measured in order to guarantee the process stability and repeatability.

A pump installed on the circuit origin keep the water circulation. After quenching, the water is recuperated by means of a recycling in a closed loop after conditioning. This system is named R4 by GH.

3.12.1. R4 unit - Description

The system is specially built to work with the usual product coolant as water containing soluble polymers additives. The excess of heat is taken out from the coolant by an industrial water flow through a heat exchanger. The standard construction and main components is:



- 01: Input / output industrial water connections
- 02: Y - type filter for protection of circuit against solid elements
- 03: Pneumatic switch valve (component applied eventually)
- 04: Manometer for primary circuit verification
- 05: Heat exchanger (remark: exchanger represented on the schema is standard welded type, depending of the specific conditions or under request, this exchanger may be dismountable type)
- 06: Product cooling water flow detector.
- 07: Product cooling water thermometer (exchanger inlet side).
- 08: Pump isolating valve (outlet side).
- 09: Product cooling water thermometer (exchanger outlet side).
- 10: Elastic connector for vibrations attenuation.
- 11: Product cooling water pump.
- 12: Pump isolating valve (inlet side).
- 13: Product cooling water tank outlet.
- 14: Quench isolating valve (inlet side).
- 15: Quenching pump.
- 16: Quench isolating valve (outlet side).
- 17: Product cooling water manometer (quench side).
- 18: Product cooling water flow detector (quench side).
- 19: Quench circuit.
- 20: Product cooling water tank.

4. Technical features

4.1. Induction hardening machine

411 1. Hardening solid state generator

Model.....	500SM20
Output power.....	500 kW
Nominal frequency range.....	5-20 kHz
Input voltage	3x415V – 50 Hz
Technology	IGBT Series digital
MF / HF voltage.....	V
Energetic efficiency.....	92%

412 2. Oscillating circuit capacitors bank

Working frequency (Bells hardening).....	10 kHz
Working frequency (Stem hardening).....	8 ~ 10 kHz

413 3. Coils interface transformer/s

Model (Bells hardening).....	Toroidal
Model (Stem hardening)	GH

414 4. Coil / inductor

Type (Bells / Stem hardening)	Single shot for outer race and Scan for tulip housing hardening
Includes product cooling quench	YES
Water fastening type	Fast connector
MF/HF electric connection method	Quick mechanical connection (Clamps)

415 5. Induction system coolant

Model	R3
Industrial water required temperature.....	22° C
Industrial water required pressure	2...7 bar
Heat exchanger type	Plate type
Additional equipment	Overpressure pump; Stainless steel piping

416 6. Product coolant

Model	R4
Industrial water required temperature.....	22° C
Industrial water required pressure.....	2...7 bar
Heat exchanger type	Plate type
Additional equipment	Paper filter; Stainless steel piping

417 7. Product handling system

Operation or stations.....	Hardening
Number of stations.....	1 Bell hardening + 1 Stem hardening
Functioning principle.....	Linear transfertype
Electric supply:	3 x 415 V – 50 Hz
Pneumatic supply	5 ÷ 6 bar

418 8. Hardening machine control system

Computer Numeric Control (CNC).....	Mitsubishi, 5 axles controlled by CNC
Man - Machine Interface (MMI)	Mitsubishi
Human held unit (HMI)	Mitsubishi

4.2. Painting

According to GH Standard, as follows:

Electrical cabinet (inside/outside) and raceways	Grey RAL 7035
Electrical panel.....	Grey RAL 7035
Mechanical tooling (base and frame).....	Grey RAL 7035
Commercial and standard components.....	as per suppliers

5. Commercial offer

5.1. Delivery and guarantee terms

The present offer is in accordance with GH Induction sales gen. terms & standard GH NG 1.7.

5.2. Acceptation tests

After manufacturing, the equipment here proposed will be tested. The tests will be performed in the GH Induction India Pvt Ltd factory in Chennai(India). GH will confirm to the customer the date of equipment availability for tests. The tests will consist of:

1. Static tests (supply verification)
 - 1.1. Concordance of manufacturing to the contract YES
 - 1.2. Accordance with safety and ergonomics YES
 - 1.3. Technical documents consistency..... YES
2. Dynamic tests (running verification)
 - 2.1. Inverter and oscillating circuit..... YES
 - 2.2. Induction equipment cooling system YES
 - 2.3. Product cooling system YES
 - 2.4. Handling system YES
 - 2.5. Control and command system YES
 - 2.6. Quality process supervision system YES
3. Treatment or process verification
 - 3.1. Treatment or process accordance with specifications YES
4. Production
 - 4.1. Cadence / speed of production YES
 - 4.2. Starting time..... YES
 - 4.3. Stability of running (H hours' test) NO
 - 4.4. Change of product / reference time NO

A representative of the buyer must attend the tests together with GH personnel in order to sign the acceptance document. The client may give GH notice of renouncement and the tests will be undertaken without the presence of the client's representative. The reception documents will then be valid as an acceptance document.

This report authorizes the delivery, signed by a customer's representative and a GH's representative, will be stated with the declaration of conformity –and subsequently, the authorization for delivery- or the modifications

5.3. Guarantee

The guarantee period is 12 months and the supplier undertakes to repair or replace as quickly as possible any parts of the supplies which, before the expiry of the guarantee period are proved to be defective due to bad material, faulty design or poor workmanship. The supplier shall bear the costs of remedying the defective parts in its works. If the repair cannot be carried out in supplier's works, the customer shall bear the related costs to the extent exceeding the customary costs of transport, personnel, travelling, living, dismantling and reassembly of the defective parts.

The guarantee starts at the taking over of the supplies in the supplier's facility (**shift unlimited**). If dispatch or taking over or erection is delayed due to reasons beyond supplier's control, the guarantee period shall end not later than 18 months after supplier's notification that the supplies are ready for dispatch.

The guarantee expires prematurely if the customer or a third party undertakes inappropriate modifications or repairs or if the customer, in case of a defect, does not immediately take appropriate steps to mitigate the damage and informs the supplier of such defect.

5.4. Delivery terms following Incoterms 2010

DAP: Hyundai WiaChennai(India)

5.5. Payment terms

40 %	as advance along with the order
50 %	along with 100 % taxes and duties against proforma invoice before dispatch of the machine
10 %	on commissioning the machine at your plant.

5.6. Delivery time and penalty for late delivery

Machine delivery time 9 Months (DAP Hyundai Wia Chennai).

The delivery time starts at the moment of formalization of the contract by parties, supplier and customer, and reception of down payment if agreed.

A Hyundai-WIA parts delay for the trials in GH will correspond in a delay of the delivery time by the same period of time. If specific parts like eddy current tester or semi conductor based items Like PCB or machine bill of material delayed by supplier can result in delay in delivery which is beyond our control should not be treated as LD.

5.7. Validity of the proposal

The present proposal is valid for 60 days.

5.8. Prices – HARDENING MACHINE LONG STEM OR

(Prices does not include taxes).

Currency: INR

<u>Item</u>	<u>Designation</u>	
1	Induction hardening equipment, including:	
1.1	One (1) induction hardening power supply 500KW, series type. Including cooling water/air	Included
1.2	One (1) Capacitor bank for bells hardening	Included
1.3	One (1) Capacitor bank for stem hardening	Included
1.4	One (1) toroidal output transformer	Included
1.5	One (1) output transformer	Included
1.6	Two (2) output plates	Included
1.7	One (1) pneumatic switch for stem and bells hardening stations	Included
2	Heat Treat handling machine	Included
2.1	One (1) hardening linear transfer type as described on quotation.	Included
2.2	One (1) set of safeties covers and protections	Included
2.3	One (1) common platform (skid format)	Included

3	Control & Command system	
3.1	One (1) Control & command system based upon CNC mitsubishi with control panel and process parameters control and supervision and hand held unit.	Included
3.2	One (1) PMS graphic parameter recorder	Included
3.3	One (1) cooling exchanger water/air for electric components.	Included
3.4	One (1) Data & statistics analysis & recording system (Data logger)	Included
3.5	One (1) Teleservice system for remote assistance by means of VPS	Included
4	Pre-acceptance tests at GH Induction facility	Included
5	Delivery and Installation	Included
5.1	Dismantling for transport and packaging	Included
5.2	Final acceptance tests on customer's site	Included
6	Documentation	Included
6.1	Five (5) paper set and e-copy of documentation in English.	Included

TOTAL Price for item no 1 to 6: DAP (Hyundai Wia-Chennai) for item no 1 to 6 for supply of one number Long stem OR machine	Rs.3,58,00,000/-
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5.9. Prices– COOLING AND ACCESORIES

Item	Designation	Included
7	Induction components equipment cooling system GH-R3 type with heat exchanger and overpressure pump for inductors and stainless steel piping.	Included
8	Quench liquid circuit system GH-R4 type with heat exchangers, heating resistances, stainless steel tank and stainless steel piping. It includes control and command system and paper filter.	Included
9	Magnetic separator on R4 circuit.	Included
Total price for item no:7 to 9 DAP HWIA Chennai for supply of one long stem outer race and tripod housing machine		Rs:72,00,000/-
Item	Designation	Included
10	Input and Output accumulator conveyor for the part capacity of 200 nos.	Included
11	Washing Station and laser marking machine and station	Included
12	Eddy current testing station with master and test coil and Dot Matrix station.	Included
13	Crate packing.	Included with machine
14	Start up, installation and training of staff Hardening machine.	Included with machine
15	Service offer pack (Training pack and 1 week production assistance).	Included with machine
Total price for item no:10 to 15 DAP HWIA Chennai for supply of one long stem outer race and tripod housing machine		1,83,60,125/-

TOTAL Price for item no 1 to 15: DAP (Hyundai Wia-Chennai) for supply of One long stem outer race and tripod housing machine	Rs. 6,13,60,125/-
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5.10. Prices– INDUCTORS AND TOOLS FOR OUTER RACE AND TULIP HOUSING

NO.	Item	Contents	Qty	Remark	Price in Rs	
16	Coil	STEM COIL (6 types)	2SET per type	TOTAL 12EA	1850000	within the supply scope for one machine
17		BELL COIL (7 types)	2SET per type	TOTAL 14EA	2715789	within the supply scope for one machine
18	JIG	Consumable Jigs for Line Change	1SET per type	TOTAL 6 SET	224737	for one machine
19		JIG (common for all types)	1SET			for one machine
20	Eddy	STEM eddy coil(male)	1SET	2EA/1SET	5927956	for one machine
21	coil	BELL eddy coil(7types)	1SET per type	2EA/1SET, Total 14EA		within the supply scope for one machine
TOTAL Price for item no 16 to 21: DAP (Hyundai Wia-Chennai) for supply of Tooling for one long stem outer race and tripod housing machine					Rs.1,07,18,482/-	

TOTAL Price for item no 1 to 21: DAP (Hyundai Wia-Chennai) for supply of one long stem outer race and tripod housing machine	Rs. 7,20,78,607/-
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TOTAL discount Price for item no 1 to 21: DAP (Hyundai Wia-Chennai)for supply of one long stem outer race and tripod housing machine – Discount @ 37.16%	Rs. 4,53,01,404/-
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GST tax extra @18%